

Function Operations, Composition & Inverses

SUPPORTING SKILLS LEGEND

Student learning objectives for this unit

I — Introduce

D — Deepen

A — Apply

R — Reinforce

- ☐ 1. I can **determine** whether a function is even, odd, or neither.

RELATED SKILLS

D Test a formula or graph for even or odd symmetry.

- ☐ 2. I can **add, subtract, multiply, and divide** functions and state the resulting domain.

RELATED SKILLS

I Add, subtract, multiply, and divide function outputs.

I Determine the domain of a sum, difference, product, quotient, or composition.

A Simplify an algebraic expression using operation properties.

- ☐ 3. I can **evaluate** a composition from formulas, graphs, or tables.

RELATED SKILLS

D Follow intermediate values to compose functions from tables.

D Substitute one function formula into another.

A Identify input and output values in a formula, graph, table, or context.

R Read and write notation for function composition and inverse functions.

- ☐ 4. I can **write** a formula for the composition of two functions and determine its domain.

RELATED SKILLS

D Determine the domain of a sum, difference, product, quotient, or composition.

D Substitute one function formula into another.

- ☐ 5. I can **decompose** a function into a composition of simpler functions.

RELATED SKILLS

I Identify simpler inner and outer functions in a composition.

A Substitute one function formula into another.

- ☐ 6. I can **determine** whether a function has an inverse on a given domain.

RELATED SKILLS

D Apply the horizontal line test for invertibility.

R Test whether a function is one-to-one on a domain.

☐ **7. I can **find and verify** an inverse function algebraically.**

RELATED SKILLS

- ☐ D Exchange variables and solve to obtain an inverse formula.
 - ☐ D Verify inverse functions using composition.
 - ☒ A Substitute one function formula into another.
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☐ **8. I can **interpret** the relationship between a function and its inverse from graphs or tables.**

RELATED SKILLS

- ☐ D Exchange domain and range between a function and its inverse.
 - ☐ D Exchange input and output values when interpreting an inverse.
 - ☐ D Reverse input and output units when interpreting an inverse.
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